

UNIVERSITI TEKNOLOGI MARA

**CIRCA: CIRCUIT INSPECTION AND
RECOGNITION USING
CONVOLUTIONAL ARCHITECTURES**

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**BACHELOR OF INFORMATION SYSTEMS
(Hons.) INTELLIGENT SYSTEMS
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**Thesis submitted in fulfilment of the requirements
for Bachelor of Information Technology (Hons.)
Intelligent Systems Engineering
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SUPERVISOR APPROVAL

CIRCA: CIRCUIT INSPECTION AND RECOGNITION USING CONVOLUTIONAL ARCHITECTURES

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This thesis was prepared under the supervision of the project supervisor, Puan Farahnatasyah Binti Abdul Hannan. It was submitted to the Faculty of Computer Science and Mathematics and was accepted in partial fulfilment of the requirements for degree of Bachelor of Information Technology (Hons.) Intelligent Systems Engineering.

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JANUARY 23, 2026

STUDENT DECLARATION

I certify that this thesis and the project to which it relates is the product of my own work and that any idea or quotation from the work of other people, published or otherwise are fully acknowledged in accordance with the standard referring practices of the discipline.

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JANUARY 23, 2026

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ABSTRACT

CIRCA (Circuit Inspection and Recognition using Convolutional Architectures) addresses the growing difficulty of diagnosing faults on modern printed circuit boards (PCBs), where extreme miniaturization, dense layouts, and technician fatigue make manual inspection slow, inconsistent, and error prone. The project is motivated by the needs of electronics repair shops, which lack access to expensive factory-grade Automated Optical Inspection systems but still require fast, reliable defect diagnosis to reduce return repairs and improve customer satisfaction. The proposed solution is an AI-driven visual inspection system that leverages convolutional neural networks, specifically YOLOv12 variants, to automatically detect and localize surface-level PCB defects such as solder bridges, missing or misaligned components, damaged traces, and burnt areas from standard camera images. Methodologically, the work follows a structured multi-phase approach: literature review and defect taxonomy formulation; dataset selection and augmentation using public PCB defect datasets; development and training of YOLOv12 Nano, Small, and Medium models; and design of a three-layer system architecture comprising an image acquisition module, a detection engine, and a technician-oriented desktop interface. The actual work performed includes constructing a repair-context defect taxonomy, curating and augmenting PCB image data, implementing and benchmarking multiple YOLOv12 configurations using metrics such as mAP, precision, recall, and inference time, and integrating the best-performing model into a standalone prototype application. The findings are expected to demonstrate that a lightweight, locally deployable CNN-based system can achieve near-industrial detection accuracy with significantly reduced diagnostic time on standard hardware, thereby offering a novel, practical, and cost-effective defect inspection solution tailored to electronics repair environments.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
AOI	Automated Optical Inspection
AQC	Automated Quality Control
AVI	Automated Vision Inspection
BOM	Bill of Materials
CIRCA	Circuit Inspection and Recognition using Convolutional Architectures
CIoU	Complete Intersection over Union
CMM	Coordinate Measuring Machine
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CPU	Central Processing Unit
DFL	Distribution Focal Loss
EfficientDet	Efficient Detection
Faster R-CNN	Faster Region-based Convolutional Neural Network
FPN	Feature Pyramid Network
FPS	Frames Per Second
GFLOPs	Giga Floating Point Operations Per Second
GPU	Graphics Processing Unit
HRIPCB	High-Resolution Images of Printed Circuit Boards
ICT	In-Circuit Testing
IoU	Intersection over Union
IR	Infrared
LCD	Liquid Crystal Display
LED	Light-Emitting Diode
mAP	mean Average Precision

ML	Machine Learning
MobileNet	Mobile Neural Network
MRI	Magnetic Resonance Imaging
MVI	Machine Vision Inspection
NLP	Natural Language Processing
PCB	Printed Circuit Board
PCBA	Printed Circuit Board Assembly
PKU	Peking University
QA	Quality Assurance
R-ELAN	Residual Efficient Layer Aggregation Network
ResNet	Residual Network
ReLU	Rectified Linear Unit
RGB	Red, Green and Blue
ROI	Return on Investment / Region of Interest
SI	International System of Units
SMT	Surface-Mount Technology
SPC	Statistical Process Control
SPPF	Spatial Pyramid Pooling Fast
SSD	Single Shot Multibox Detector
UV	Ultraviolet
YOLO	You Only Look Once
YOLOv5	You Only Look Once version 5
YOLOv7	You Only Look Once version 7
YOLOv8	You Only Look Once version 8
YOLOv12	You Only Look Once version 12
2D	Two-Dimensional
3D	Three-Dimensional

CHAPTER 1

INTRODUCTION

This chapter outlines the motivation, problem statement, and goals of the CIRCA project. It introduces the challenges of manual PCB defect inspection, states the research questions and objectives, and summarizes the scope and significance of the proposed solution.

1.1 Research Background

The rapid advancement of electronic technology has led to an unprecedented increase in the complexity and miniaturization of electronic devices, with Printed Circuit Boards (PCBs) serving as the fundamental backbone of modern electronics (Bhattacharya & Cloutier, 2022). PCBs are ubiquitous components found in virtually every electronic device, ranging from smartphones and laptops to medical equipment and automotive systems (Wang et al., 2023). As the electronics industry continues to evolve, the demand for high-quality, defect-free PCBs has intensified significantly, particularly given that even minor defects can compromise device functionality, reliability, and safety (Klco et al., 2023).

In contemporary electronics manufacturing, quality control has emerged as a critical factor in ensuring product reliability and customer satisfaction. Traditional manual inspection methods, while historically prevalent, have increasingly proven inadequate in addressing the challenges posed by modern PCB complexity (Aggarwal et al., 2022). The miniaturization of electronic components, coupled with the increasing density of circuit layouts, has rendered visual inspection by human operators time-consuming, error-prone, and often

ineffective in detecting subtle defects (Bhanumathy et al., 2021). Studies have demonstrated that manual inspection methods can exhibit error rates ranging from 10% to 20%, particularly during extended work shifts where human fatigue becomes a significant factor (Law et al., 2024).

The limitations of manual inspection have catalysed the development and adoption of automated inspection systems leveraging machine learning and computer vision technologies (Anh Nguyen et al., 2024). Automated Optical Inspection (AOI) systems have gained widespread acceptance in high-volume manufacturing environments, utilizing advanced imaging techniques and sophisticated algorithms to detect defects with unprecedented accuracy and speed. Recent advancements in deep learning, particularly Convolutional Neural Networks (CNNs), have revolutionized PCB defect detection by enabling systems to automatically learn and identify complex defect patterns without extensive manual feature engineering (Bhattacharya & Cloutier, 2022).

Contemporary research has demonstrated that deep learning-based approaches, including architectures such as YOLO (You Only Look Once), ResNet, and EfficientDet, can achieve defect detection accuracies exceeding 95% while maintaining real-time processing capabilities (Law et al., 2024; Yang & Yu, 2024). These systems have proven particularly effective in identifying various defect types, including solder bridges, missing components, misaligned parts, open circuits, and spurious copper deposits. The integration of artificial intelligence in PCB inspection represents a paradigm shift from reactive quality control to proactive defect prevention, enabling manufacturers to identify and rectify issues early in the production process (Ghelani, 2024).

Within the context of electronics repair and maintenance, the application of automated defect detection systems remains relatively unexplored. Computer and mobile phone repair technicians face considerable challenges in diagnosing PCB-related faults, particularly given the increasing miniaturization of components and the growing complexity of circuit designs. The manual diagnosis process is not only time-consuming but also highly dependent on the technician's expertise and experience, leading to inconsistencies in repair quality and extended turnaround times. Furthermore, undetected defects can result in

subsequent device failures, customer dissatisfaction, and potentially hazardous situations in critical applications.

This research addresses this identified gap by proposing CIRCA (Circuit Inspection and Recognition using Convolutional Architectures), an automated PCB defect detection system specifically designed to assist repair technicians in rapidly and accurately identifying circuit board defects. By leveraging state-of-the-art machine learning techniques, particularly convolutional neural networks, the proposed system aims to augment the capabilities of repair professionals, reduce diagnostic time, minimize human error, and ultimately improve the quality and efficiency of electronics repair services.

1.2 Problem Statement

The electronics repair industry faces several interconnected challenges that significantly impact service quality, operational efficiency, and customer satisfaction. These challenges stem primarily from the limitations inherent in current manual inspection methodologies and the increasing complexity of modern electronic devices.

Firstly, the detection of defects in modern PCBs through manual visual inspection has become increasingly problematic due to progressive component miniaturization and elevated circuit density (Djuberh et al., 2025). Contemporary PCBs frequently incorporate surface-mount technology (SMT) components measuring less than 1 millimetre, fine-pitch integrated circuits with pin spacing below 0.5 millimetres, and multi-layer board constructions with internal connections invisible to surface inspection (Adibhatla et al., 2020). Under these circumstances, even experienced technicians equipped with magnification tools struggle to identify subtle defects such as hairline cracks, micro-solder bridges, or incipient component failures.

Secondly, manual inspection processes are inherently time-consuming and subject to human limitations. A comprehensive visual examination of a complex PCB can require 10 to 30 minutes or more, depending on board complexity and the number of components present. This extended inspection time directly

translates to increased repair costs, longer customer wait times, and reduced throughput for repair facilities. Moreover, human inspectors are susceptible to fatigue, distraction, and subjective judgment variations, which can lead to missed defects or false positives. Research indicates that inspector accuracy declines significantly after extended periods of continuous inspection, with defect detection rates dropping by up to 30% after four hours of sustained work.

Thirdly, the reliability and consistency of manual inspection are heavily dependent on individual technician expertise and experience levels (Sharma et al., 2024). Novice technicians may lack the knowledge to recognize certain defect types or understand their implications, while even experienced professionals may overlook defects in unfamiliar board designs or when confronting novel failure modes. This variability in inspection quality creates inconsistencies in service delivery and can erode customer confidence in repair services.

Furthermore, certain categories of PCB defects present challenges for visual detection. Internal solder joint failures, delamination within multi-layer boards, intermittent connection problems, and incipient component degradation may not manifest obvious visual indicators during static inspection (Hu & Wang, 2020). These "hidden" defects can evade detection during manual inspection yet subsequently cause device malfunction, necessitating repeated repair attempts and generating customer dissatisfaction.

The consequences of undetected or misdiagnosed PCB defects extend beyond immediate repair failures. In critical applications such as medical devices, automotive electronics, or industrial control systems, undetected defects can pose serious safety hazards or result in catastrophic system failures (Liao et al., 2021). Even in consumer electronics, unresolved PCB issues can lead to premature device failure, data loss, fire hazards from electrical shorts, or exposure to hazardous materials from damaged battery management circuits.

From an economic perspective, inefficient defect detection processes impose substantial costs on repair businesses. Extended diagnostic times reduce shop productivity and limit the number of repairs that can be completed daily. Misdiagnosed problems result in unnecessary component replacements, wasted

parts inventory, and repeated repair attempts. Failed repairs damage business reputation and may result in warranty claims or legal liabilities. These factors collectively undermine the profitability and sustainability of electronics repair operations.

Current automated inspection technologies, while highly effective in manufacturing environments, are generally designed for high-volume production scenarios and may not be optimally configured for the diverse range of devices and repair contexts encountered in service centres (Huang et al., 2023). Existing AOI systems are often expensive, require specialized training to operate, and may lack the flexibility to accommodate the variety of PCB types and device configurations typical of repair workflows. Consequently, there exists a clear need for accessible, cost-effective, and adaptable automated inspection solutions tailored specifically to the requirements of electronics repair and service operations.

This research therefore addresses the following core problem: *How can machine learning-based automated defect detection systems be effectively designed and implemented to enhance the speed, accuracy, and reliability of PCB fault diagnosis in electronics repair environments, thereby improving service quality and operational efficiency while minimizing the impact of human error and expertise variability?*

1.3 Research Questions

To comprehensively address the identified problem and guide the research investigation, this study seeks to answer the following research questions:

RQ1: How to identify and categorize common PCB defect types based on their visual characteristics for automated detection in electronics repair contexts?

RQ2: How to design and develop CNN-based models (specifically YOLOv12 variants) optimized for accurate and efficient PCB defect detection in repair service environments?

RQ3: How to develop and evaluate a user-friendly PCB defect detection prototype (CIRCA system) for practical deployment in electronics repair facilities?

1.4 Research Objectives

Based on the research questions formulated above, this study establishes the following specific objectives:

RO1: To identify, categorize, and document the most prevalent PCB defect types encountered in electronics repair based on their characteristic visual signatures suitable for automated detection.

RO2: To design and develop CNN-based detection models using YOLOv12 architecture variants (Nano, Small, Medium), conducting comparative performance evaluation to identify the optimal model for PCB defect detection in repair contexts.

RO3: To develop the CIRCA (Circuit Inspection and Recognition using Convolutional Architectures) system prototype with a user-friendly interface and evaluate its performance through empirical testing using benchmark metrics (precision, recall, F1-score, and mAP).

1.5 Research Scope

To ensure focused and achievable research outcomes, this study establishes the following scope boundaries:

1.5.1 Technical Scope

The research focuses specifically on visual defect detection in PCBs using optical imaging and convolutional neural network analysis. The study encompasses the detection and classification of surface-visible defects that can be identified through optical inspection, including but not limited to solder bridges, cold solder joints, missing components, misaligned components, damaged traces, burnt components, and physical board damage. The research

does not extend to electrical testing, functional verification, or the detection of defects requiring X-ray, infrared, or other specialized imaging modalities beyond standard optical photography.

The technical implementation focuses on deep learning approaches, specifically convolutional neural networks and modern object detection architectures such as YOLO, ResNet, and related variants. While the research reviews traditional computer vision techniques for context, the primary emphasis remains on learning-based methods that have demonstrated superior performance in contemporary literature. The study develops a standalone software prototype rather than a fully integrated production system, prioritizing proof-of-concept demonstration and validation of core detection capabilities.

1.5.2 Device Scope

The research concentrates on PCBs commonly found in consumer electronics devices, particularly those frequently encountered in repair shops, including smartphones, laptops, tablets, and similar portable electronic devices. These device categories represent the majority of repair shop workload and provide a manageable yet representative scope for system development and validation. The study does not address specialized industrial electronics, aerospace systems, medical devices, or other domain-specific applications that may require different detection criteria or certification requirements.

1.5.3 Defect Scope

The investigation focuses on manufacturing defects, physical damage, and wear-related failures that manifest visually on PCB surfaces. Priority is given to defect types that are both common in repair contexts and amenable to optical detection, as identified through literature review and consultation with repair professionals. The research does not address intermittent faults, software-related issues, or defects that can only be detected through electrical testing or operational verification.

1.5.4 Operational Scope

The system is designed for use by repair technicians in small to medium-sized electronics repair facilities, rather than high-volume manufacturing environments. The research assumes availability of standard digital cameras or smartphone cameras for image capture, rather than specialized industrial imaging equipment. The computational requirements target standard desktop or laptop computers commonly available in repair shops, rather than requiring specialized AI hardware accelerators or cloud-based processing infrastructure.

1.5.5 Geographical and Temporal Scope

The research is conducted within the academic environment of Universiti Teknologi MARA during the specified project timeline. Data collection, model development, and system validation activities are constrained by the resources, facilities, and time available within the university's Final Year Project framework. Pilot deployment and testing are conducted in collaboration with local electronics repair facilities within accessible proximity to the university.

1.5.6 Limitations and Exclusions

The research does not attempt to develop a commercially deployable product with full production-level features, extensive user interface polish, or comprehensive customer support mechanisms. Security, privacy, and data protection considerations are addressed at a fundamental level appropriate for an academic prototype but do not extend to enterprise-grade security implementations. The study does not include longitudinal performance monitoring beyond the project duration, though recommendations for such monitoring are provided in the conclusions.

The research recognizes that while the CIRCA system aims to assist repair technicians, it is not intended to completely replace human expertise and judgment. The system serves as a diagnostic aid and decision support tool rather than an autonomous repair solution. Final repair decisions, component selection, and quality verification remain the responsibility of qualified technicians.

1.6 Significance of Study

This research makes several important contributions to both academic knowledge and practical application in the fields of computer vision, machine learning, and electronics repair services.

1.6.1 Academic Significance

From a research perspective, this study advances the application of deep learning techniques to a relatively underexplored domain---electronics repair diagnostics. While substantial research exists on automated defect detection in manufacturing contexts, the adaptation of these technologies to repair service environments presents unique challenges and opportunities that have received limited scholarly attention. The research contributes to the theoretical understanding of how convolutional neural networks can be optimized for diverse PCB types and defect categories encountered in repair scenarios, as opposed to the more controlled manufacturing environments typically addressed in existing literature.

The development of a PCB defect dataset specifically tailored to repair contexts represents a valuable contribution to the research community. Most existing PCB defect datasets focus on manufacturing defects in controlled production environments, whereas repair scenarios involve a broader range of failure modes, device types, and image capture conditions. The annotated dataset generated through this research, if made publicly available, could facilitate future research by other scholars investigating similar applications.

The comparative analysis of different CNN architectures for PCB defect detection in resource-constrained environments provides practical insights for researchers working on edge computing applications, mobile machine learning deployment, and real-time computer vision systems. The findings regarding model optimization, inference speed versus accuracy trade-offs, and resource utilization have applicability beyond PCB inspection to other domains requiring efficient computer vision solutions.

1.6.2 Practical Significance

For electronics repair businesses, this research offers a pathway to significantly enhance service quality, operational efficiency, and competitive positioning. The CIRCA system has the potential to reduce diagnostic time by up to 70%, based on benchmarks from similar automated inspection implementations in manufacturing contexts. This time savings translates directly to increased repair throughput, reduced customer wait times, and improved shop profitability. For a typical repair facility completing 20-30 repairs daily, even a 30% reduction in diagnostic time could enable an additional 6-9 repairs per day, representing substantial revenue increase.

The improvement in defect detection accuracy addresses a critical pain point in repair operations---misdiagnosis and return repairs. Industry data suggests that 15-25% of electronics repairs result in return visits due to incomplete diagnosis or missed secondary faults. By providing consistent, comprehensive defect detection, the CIRCA system can potentially reduce return repairs by identifying all present defects during initial diagnosis, thereby improving first-time fix rates and customer satisfaction.

For repair technicians, particularly those with limited experience, the system serves as an educational tool and decision support aid. Novice technicians can leverage the system's knowledge to identify defects they might otherwise miss, accelerating their learning curve and improving diagnostic confidence. The system also provides documentation capabilities, allowing technicians to capture and annotate defect images for customer communication, quality records, and knowledge sharing among team members.

1.6.3 Economic Significance

The implementation of automated defect detection has important economic implications for the electronics repair industry. By reducing dependence on highly experienced (and expensive) technicians for routine diagnostics, repair shops can optimize their staffing models and reduce labour costs. The system

enables junior technicians to handle more complex diagnostics with supervisory oversight, rather than requiring senior technician involvement in every diagnosis.

The reduction in misdiagnosis and unnecessary component replacements represents direct cost savings through reduced parts wastage and more efficient inventory management. Furthermore, improved first-time fix rates reduce warranty costs and the resources required for rework. For larger repair chains, these savings can accumulate to significant annual cost reductions while simultaneously improving customer satisfaction and retention.

1.6.4 Environmental Significance

From a sustainability perspective, improved repair diagnostics contribute to the circular economy by extending device lifespans and reducing electronic waste. Accurate defect detection enables more effective repairs, preventing premature device replacement and the associated environmental impacts of manufacturing new devices and disposing of old ones. By making repairs more reliable and efficient, the CIRCA system can help make device repair more economically attractive relative to replacement, encouraging consumers to choose repair over disposal.

1.6.5 Social Significance

The democratization of advanced diagnostic capabilities through accessible, affordable technology has social implications for small businesses and independent repair professionals. By reducing the technical barriers and capital requirements for high-quality diagnostics, the system can help level the playing field between small independent repair shops and larger service chains, supporting local businesses and employment.

For consumers, improved repair reliability and reduced turnaround times enhance the viability of repair as an alternative to replacement, promoting more sustainable consumption patterns. In regions where access to new electronics is limited by economic constraints, effective repair services supported by tools like CIRCA can help bridge digital divides and extend technology access.

1.6.6 Educational Significance

Within the context of Universiti Teknologi MARA's Intelligent Systems Engineering program, this research provides valuable educational outcomes. The project demonstrates the practical application of machine learning concepts to real-world problems, bridging the gap between theoretical knowledge and practical implementation. The interdisciplinary nature of the research---combining computer vision, machine learning, electronics knowledge, and user interface design---provides comprehensive learning experiences that prepare students for professional practice in emerging technology fields.

1.7 Expected Solution

The CIRCA (Circuit Inspection and Recognition using Convolutional Architectures) system represents an integrated hardware and software solution designed to automate PCB defect detection in electronics repair environments. The expected solution encompasses several key components and capabilities that collectively address the identified research problem.

1.7.1 System Architecture

The CIRCA system architecture consists of three primary subsystems working in concert: an image acquisition module, a defect detection engine, and a user interface layer.

The **image acquisition module** enables technicians to capture high-resolution images of PCBs using readily available imaging devices such as digital cameras, webcams, or smartphone cameras. The module incorporates automated image quality assessment to ensure captured images meet minimum standards for successful detection, providing real-time feedback to guide technicians in optimal image capture. Image preprocessing capabilities include automatic lighting correction, perspective transformation for images captured at angles, and noise reduction to enhance detection accuracy.

The **defect detection engine** forms the core of the system, implementing trained convolutional neural network models for automated defect identification and classification. The engine is designed to process images in real-time or near-real-time, providing detection results within 5-10 seconds of image capture on standard computer hardware. The system employs ensemble methods combining multiple detection models to maximize accuracy and minimize false positives, with confidence scoring to indicate detection certainty for each identified defect.

The **user interface layer** provides an intuitive, user-friendly interface requiring minimal technical expertise to operate. The interface displays detection results with visual highlighting of identified defects, classification labels, and confidence scores. It includes features for manual verification and correction of detection results, allowing technicians to confirm or refute automated findings. The system maintains a database of inspection results for historical reference, quality tracking, and technician training purposes.

1.7.2 Core Functionalities

The CIRCA system delivers the following essential functionalities:

- **Automated Defect Detection:** The system automatically identifies and localizes common PCB defects including solder bridges, cold solder joints, missing components, misaligned components, damaged traces, burnt or discoloured components, physical board damage such as cracks or delamination, and contamination or corrosion. Detection operates on full PCB images as well as zoomed regions of interest for detailed examination.
- **Defect Classification:** Beyond detection, the system classifies identified defects into specific categories, providing technicians with actionable diagnostic information. Classification includes defect type identification, severity assessment (critical, major, minor), and location specification with coordinate or component-level granularity.
- **Result Visualization:** Detection results are presented through an intuitive visual interface with defect bounding boxes overlaid on PCB images, color-

coded severity indicators, confidence scores for each detection, and detailed defect descriptions and repair recommendations where applicable.

- **Reporting and Documentation:** The system generates comprehensive inspection reports documenting detected defects, capturing timestamp and technician information, and providing exportable formats for customer communication and quality records. The reporting capability supports workflow integration and quality management requirements.
- **Continuous Learning:** The system architecture supports ongoing model improvement through collection of validation data from technician confirmations or corrections, periodic model retraining incorporating new defect examples, and performance monitoring to track detection accuracy over time.

1.7.3 Performance Expectations

Based on contemporary research in PCB defect detection and the capabilities of state-of-the-art convolutional neural networks, the CIRCA system is expected to achieve the following performance benchmarks:

- **Detection Accuracy:** Overall detection accuracy (mAP@0.5) of at least 90% across all defect categories, with precision exceeding 85% to minimize false positives and recall exceeding 85% to minimize missed defects. For critical defect types such as solder bridges and shorts, the system targets detection rates above 95% given their importance for device safety and functionality.
- **Processing Speed:** Image processing and defect detection completed within 10 seconds on standard computing hardware, enabling real-time or near-real-time feedback during diagnosis. This performance target ensures the system integrates seamlessly into existing repair workflows without creating bottlenecks.
- **Usability:** The system interface is designed for operation by technicians with basic computer literacy but without machine learning expertise, requiring no more than 30 minutes of training for proficient use. The

interface supports rapid image capture and processing with minimal user interaction required.

- **Reliability:** The system demonstrates consistent performance across diverse PCB types, device categories, and imaging conditions encountered in repair contexts. Error handling mechanisms gracefully manage edge cases, poor-quality images, or unexpected inputs without system crashes or data loss.

1.7.4 Technical Implementation

From a technical perspective, the CIRCA solution employs proven deep learning architectures with documented effectiveness in object detection tasks. The research investigates and compares multiple architectural approaches including:

- **YOLO** (You Only Look Once) variants, specifically YOLOv12 (Nano, Small, Medium) offering excellent speed-accuracy trade-offs suitable for real-time applications. YOLOv12's attention-centric design and R-ELAN backbone have demonstrated strength in detecting small objects—a critical requirement for miniaturized PCB components.
- **ResNet-based architectures** providing robust feature extraction through residual learning mechanisms, enabling the detection of subtle defects that might be missed by simpler networks.
- **EfficientDet and related models** offering parameter-efficient architectures suitable for deployment on resource-constrained hardware while maintaining high detection accuracy.

The selected architecture(s) are trained on a carefully curated dataset of PCB images representing diverse defect types, device categories, and imaging conditions. Data augmentation techniques expand the effective dataset size and improve model generalization. Transfer learning from models pre-trained on large-scale image datasets (such as COCO or ImageNet) accelerates training and improves performance, particularly when labelled PCB data is limited.

The implementation employs modern deep learning frameworks such as PyTorch or TensorFlow, providing flexibility, extensive community support, and efficient model development workflows. Model optimization techniques including quantization, pruning, and knowledge distillation are investigated to

reduce model size and improve inference speed without significant accuracy degradation.

1.7.5 Deployment Considerations

The CIRCA system is designed with practical deployment requirements in mind. Installation requires only standard computing hardware (mid-range desktop or laptop computer with dedicated graphics capabilities recommended but not mandatory), readily available imaging devices, and the CIRCA software package with pre-trained models. The system operates standalone without requiring constant internet connectivity, addressing potential concerns about data privacy and network reliability in repair shop environments.

Documentation includes user manuals for repair technicians, technical guides for system administrators, and troubleshooting resources for common issues. Training materials comprising video tutorials and hands-on workshops facilitate rapid user adoption and proficiency development.

1.7.6 Expected Impact

Upon successful implementation, the CIRCA system is expected to deliver measurable improvements across multiple dimensions of repair service quality and efficiency. Diagnostic time reduction of 40-70% relative to manual inspection alone increases repair throughput and reduces customer wait times. Defect detection accuracy improvement of 15-30% compared to manual inspection by average technicians reduces misdiagnosis rates and return repairs. Consistency enhancement providing uniform diagnostic quality regardless of individual technician experience levels standardizes service quality across the organization. Documentation improvement through automated capture and archiving of defect images and inspection results supports quality management, warranty claims, and customer communication.

By achieving these outcomes, the CIRCA system addresses the core research problem of enhancing PCB fault diagnosis in electronics repair environments

while providing a practical, accessible solution that can be adopted by repair facilities of various sizes and resource levels.

CHAPTER 2

LITERATURE REVIEW

This chapter surveys prior work on PCB defect detection and deep learning-based visual inspection. It compares existing methods, datasets, and YOLO-based models, and identifies the research gap that CIRCA aims to fill.

2.1 Mind Map

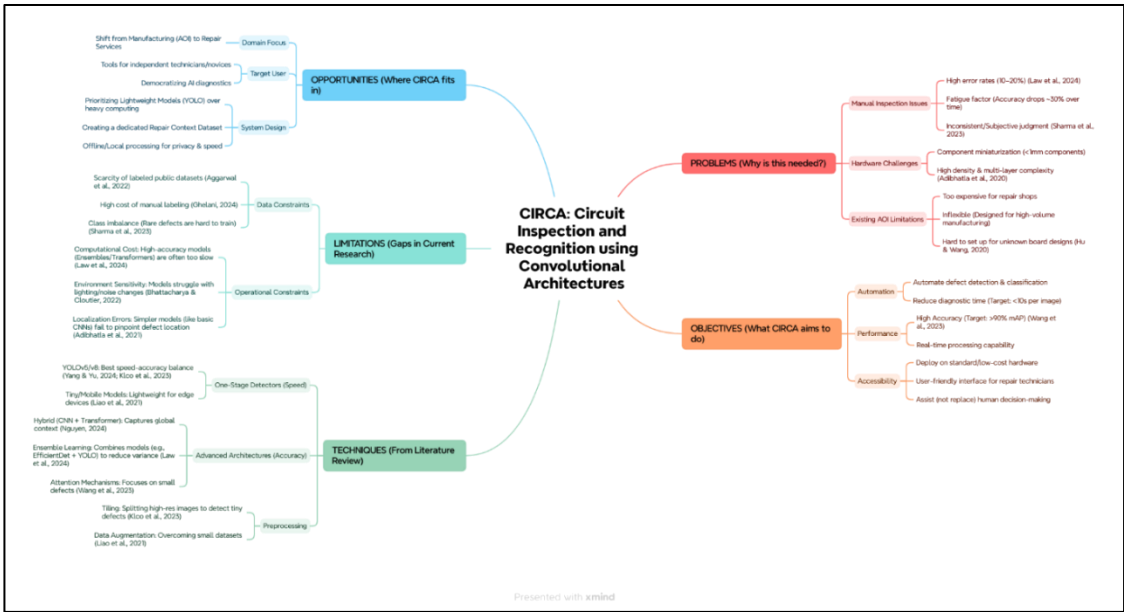


Figure 1: Mind Map

2.2 Literature Review Table

Author (Year)	Title	Objective	Technique	Data	Results	Limitation	Notes/Remarks
(Djuberh et al., 2025)	Automated Inspection of Printed Circuit Board Assemblies	Automate incoming inspection using segmentation masks instead of bounding boxes.	DeepLabv3+ (Semantic Segmentation) with ResNet101 encoder and Spatial Pyramid Pooling.	Unibap Space Solutions internal dataset (2057 images).	F1 Score: ~0.6; Dice Loss: ~0.1.	Model performance not adequate for production; struggles with complex scenes.	Thesis project. Suggests object detection might be more viable than segmentation for this task.
Yang & Yu (2024)	Deep Learning Based Defect Detection Research on Printed Circuit Boards	Achieve high-speed, high-accuracy detection for industrial lines.	YOLOv8 enhanced with C2f module and Spatial Pyramid Pooling Fast (SPPF)	PKU Open Lab Dataset (11,361 images, 6 defect types).	mAP: 92.3%; Speed: 157.2 FPS.	Relies on a single model architecture (potential robustness issues).	Validates YOLOv8's superior balance of speed/accuracy over v5 and v7.
Law et al. (2024)	Enhancing Printed Circuit Board Defect Detection Through Ensemble Learning	Increase robustness against image noise and model variance.	Ensemble Learning (EfficientDet, MobileNet SSDv2, Faster RCNN, YOLOv5) with hybrid voting.	Combined public datasets (PKU, TDD-net, etc.).	Accuracy: 95%; mAP: 80.3%.	High computational cost; inference time is the sum of all base models.	Effectively addresses "noisy" data by aggregating predictions from different architectures.

Nguyen & Nguyen (2024)	Towards High Quality PCB Defect Detection Leveraging State-of-the-Art Hybrid Models	Detect subtle, small-scale defects with high precision.	Hybrid CNN-Transformer (ResNet + Bottleneck Transformer), Ghost Convolution, FCOS Head.	PKU Dataset (Augmented to 5,814 samples).	mAP: 99.2%; FPS: 51; GFLOPs: 41.0.	Slightly lower FPS compared to pure CNN lightweight models.	Uses Wise-IoU loss to focus on high-overlap regions; Transformer captures global context.
Ghelani (2024)	AI-Driven Quality Control in PCB Manufacturing: Enhancing Production Efficiency and Precision	Quantify the operational impact of AI integration in manufacturing.	Case Study implementing Machine Learning & Computer Vision (general).	Company X production line data.	Defect detection rate increased from 92% to 99.5%.	High initial setup cost (\$1.5M); requires massive data labelling.	Validates economic feasibility; inspection time reduced from 20s to 5s.
Huang et al. (2023)	Neighbourhood Correlation Enhancement Network for PCB Defect Classification	Distinguish "True" (functional) vs. "Pseudo" (cosmetic) defects using context.	NCE-Net with Relevance Residual Blocks (RRB) to analyse defect surroundings.	PCB-2-DET (Custom: 6400 images labelled true/pseudo).	Accuracy: ~87.5%; High specificity for pseudo defects.	Parameter-heavy compared to lightweight models; requires specific "true/pseudo" labels.	Introduces Specific Integration (SI) to filter fuzzy data; critical for reducing false alarm rates.

Wang et al. (2023)	Toward surface defect detection in electronics manufacturing by an accurate and lightweight YOLO-style object detector	Process high-resolution images in real-time with global feature extraction.	ATT-YOLO (YOLOv5-small + one Self-Attention module + Improved Auto-Anchor).	LCFC-Laptop (14,478 defects) & COCO dataset.	mAP0.5: 92.8% (LCFC); FPS: 111 (TensorRT).	Dataset contains some noisy/mislabelled samples.	Demonstrates that a single attention module improves accuracy without heavy computational cost.
Klco et al. (2023)	Quality inspection of specific electronic boards by deep neural networks	Detect tiny solder splashes in power modules (small area, low contrast).	YOLOv8n (nano) vs. YOLOv5 vs. Faster R-CNN.	Custom dataset (15MPix images tiled to 640x640).	mAP: 96.6%; Speed: 90ms.	Limited to a single defect class (solder splash).	YOLOv8n proved best for speed/accuracy trade-off; uses custom tiling to avoid cutting objects.
Sharma et al. (2023)	PCB Defect Detection Using Deep Learning Methods	Compare traditional Image Processing vs. Deep Learning efficacy.	YOLOv5 vs. OpenCV (Image Subtraction/Thresholding).	PKU Open Lab Dataset (693 images).	YOLOv5 F1: ~0.60; OpenCV: High detection but no classification.	OpenCV cannot classify defect types; dataset class imbalance affected DL results.	Establishes DL as superior for classification, while OpenCV is only useful for basic discrepancy spotting.

Bhattacharya & Cloutier (2022)	End-to-end deep learning framework for printed circuit board manufacturing defect classification	High accuracy on low-resolution images using attention mechanisms.	YOLOv5 + Transformer module (C3TR) at backbone/neck junction.	HRIPCB Dataset (1386 images, augmented) .	mAP: 98.1% (low-res inputs); Params: 7.02M.	Assumes test/train data have identical distribution (lighting/environment).	Model requires 3x fewer parameters than standard Faster R-CNN; high improvement on "Spur" defects.
Liao et al. (2021)	YOLOv4-MN3 for PCB Surface Defect Detection	Reduce model parameters for cost-effective deployment	YOLOv4 + MobileNetV3 backbone + Mish activation function.	Custom dataset (2008 images, augmented to 19,029).	mAP: 98.64%; FPS: 56.98; Params: 39.59M.	Supervised learning dependence on large, labelled datasets.	MobileNetV3 backbone significantly reduced computational cost (Madds) compared to CSPDarknet53.
Adibhatla et al. (2021)	Defect detection in PCBs using convolutional neural network	Automate manual inspection with a simple architecture (Conference Paper).	Basic 4-Layer CNN (Conv, ReLU, MaxPool, FC)	Linear scan CCD images (832 samples).	Accuracy: ~70% (Confusion Matrix normalized) .	Very low accuracy; simple classification only (no localization).	Serves as a baseline showing why complex architectures (YOLO/ResNet) are necessary.

Aggarwal et al. (2021)	A Survey on Automatic Printed Circuit Board Defect Detection Techniques	Review state-of-the-art detection methods (Survey Paper).	Literature Review (Machine Vision vs. Deep Learning).	N/A	N/A	Machine vision struggles with noise/lighting; DL requires massive data.	Concludes DL is more precise/adaptable but highlights the data scarcity bottleneck.
Hu & Wang (2020)	Detection of PCB Surface Defects With Improved Faster-RCNN and Feature Pyramid Network	Detect tiny defects without strict template alignment.	Faster R-CNN + ResNet50-FPN + GARP N (Guided Anchor).	Custom dataset (1500 images, augmented to 12,000).	mAP: 94.2%; Speed: 0.08s/img (~12 FPS).	Slower inference speed compared to one-stage detectors (YOLO).	GARP N reduces unnecessary anchors by 90%, improving efficiency over standard RPN.
Adibhatla et al. (2020)	Defect Detection in Printed Circuit Boards Using You-Only-Look-Once Convolutional Neural Networks	High-speed detection for industrial AOI integration (Journal Article).	Tiny-YOLO-v2 (Compact CNN).	Custom AOI dataset (11,000 images).	Accuracy: 98.79% (Batch size 32).	Higher localization errors than Region-based methods; lower recall.	Proves viability of compact models (<50MB) for embedded computer vision systems.

Table 1: Literature Review Table

2.3 Literature Review Writing

The increasing miniaturisation of Printed Circuit Boards (PCBs) has rendered traditional manual inspection inefficient and prone to error, with studies reporting error rates of 10–20% due to technician fatigue and visual limitations (Aggarwal et al., 2022; Law et al., 2024). While Automated Optical Inspection (AOI) systems based on traditional machine vision have successfully automated quality control in high-volume manufacturing, they remain prohibitively expensive and inflexible for the high-variety, low-volume environment of electronics repair shops (Ghelani, 2024; Hu & Wang, 2020). Consequently, research has shifted toward deep learning approaches, which offer superior adaptability to noise and complex defect patterns compared to rigid template-matching techniques (Bhattacharya & Cloutier, 2022).

Recent literature demonstrates that Convolutional Neural Networks (CNNs), particularly single-stage detectors like YOLO, achieve an optimal balance between speed and accuracy for PCB inspection. Studies by Wang et al. (2023), Adibhatla et al. (2020), and Liao et al. (2021) confirm that lightweight architectures (e.g., YOLOv5, MobileNet backbones) can achieve defect detection accuracy exceeding 90% with real-time inference speeds, making them suitable for resource-constrained deployments. Furthermore, advanced techniques such as ensemble learning (Law et al., 2024) and hybrid CNN-Transformer models (Anh Nguyen et al., 2024) have pushed detection precision even higher for subtle defects like tiny solder splashes, though often at a higher computational cost that may not be viable for standard repair shop hardware.

Despite these advancements, a significant gap remains most of the existing research focuses on optimizing algorithms for controlled industrial production lines rather than the heterogeneous conditions of repair services. Most datasets and models do not account for the specific needs of repair technicians, such as the requirement for low-cost hardware, offline capability, and user-friendly interfaces for non-experts (Aggarwal et al., 2022; Klco et al., 2023). The CIRCA project addresses this limitation by adapting proven YOLO-based architectures specifically for the repair context, aiming to democratise high-accuracy automated diagnostics for independent technicians.

CHAPTER 3

RESEARCH METHODOLOGY

This chapter describes the methodology used to design and evaluate CIRCA. It explains the research design, data preparation, YOLOv12 model development, system architecture, and the metrics applied to assess system performance.

3.1 Research Design

This project adopts a positivist research paradigm with a quantitative, experiment-based research design. The positivist paradigm is characterized by the belief that knowledge can be obtained through objective observations and measurements. In this study, positivism is applied through:

- **Empirical observation:** The research relies on observable PCB image data that can be directly measured and analysed.
- **Quantitative methods:** Performance is evaluated using measurable, numerical metrics such as mean Average Precision (mAP), precision, recall, and F1-score.
- **Hypothesis testing:** The study tests the hypothesis that CNN-based models can achieve superior defect detection accuracy compared to manual inspection methods.
- **Objectivity and replicability:** The experimental design ensures that results can be verified and replicated by other researchers using the same datasets and methodologies.

A deductive approach is employed, where existing literature on PCB defect detection and deep learning is reviewed, suitable architectures (YOLOv12 variants) are selected based on reported strengths, and their effectiveness is tested empirically on publicly available PCB defect datasets. This follows the

positivist principle of moving from theory to empirical testing. The study represents applied research aimed at solving a practical problem in electronics repair through systematic experimentation. The time horizon is cross-sectional, as all data collection, model development, and evaluation activities are conducted within the defined Final Year Project duration.

3.2 Research Methodology and Design

The research is organized into seven phases that guide the work from problem understanding to system delivery and documentation.

Phase 1 – Preliminary Study (O1)

- a. Conduct a comprehensive literature review on PCB defect detection, YOLO-based object detection, and challenges in electronics repair.
- b. Identify gaps such as the lack of accessible, repair-oriented automated inspection tools.

Phase 2 – Knowledge Acquisition (O2)

- a. Analyse existing PCB defect datasets and literature to identify prevalent defect types and their visual characteristics.
- b. Where possible, consult repair technicians to validate the relevance of these defect types in real repair contexts.

Phase 3 – Data Collection and Preparation (O4)

- a. Use publicly available PCB defect datasets such as HRIPCB and PCB-defect/PCB-Defect-style datasets that contain labelled images with six surface defect classes.
- b. Perform preprocessing (resizing, normalization, label conversion) and data augmentation (flips, rotations, brightness/contrast changes, mosaic/mix-up) and split the data into training, validation, and test sets.

Phase 4 – System Design (O3, O6)

- a. Design the CIRCA system architecture, consisting of an image acquisition module, a YOLOv12-based defect detection engine, and a user interface layer for technicians.

- b. Specify data flow, hardware and software requirements, and user interface structure.

Phase 5 – System Development (O3, O5, O6)

- a. Implement the CIRCA prototype by integrating the three system layers.
- b. Implement and train three YOLOv12 model variants (Nano, Small, Medium) on the prepared datasets using an appropriate deep learning framework.
- c. Develop a user-friendly interface for image upload, result visualization, and report generation.

Phase 6 – Testing and Evaluation (O5, O7, O8, O9)

- a. Evaluate each model variant using mAP, precision, recall, F1-score, and inference speed on the validation and test sets.
- b. Compare the performance of the three models and, where possible, benchmark against published results on similar datasets.
- c. If feasible, conduct a small pilot or simulated deployment in a repair environment to collect user feedback and assess practical utility.
- d. Analyse errors and document limitations and challenges.

Phase 7 – Documentation and Dissemination (O9, O10)

- a. Document the full research process, system design, implementation details, experimental results, and limitations in the thesis.
- b. Prepare technical documentation and a code repository to support reproducibility and potential future work.

3.3 Methodology Table

The following table maps each research objective to the corresponding research phase, key activities, methods/techniques, and expected deliverables. This structured approach ensures systematic achievement of all research goals.

Objective	Phase	Key Activities	Methods/Techniques	Primary Deliverable
RO1: To identify, categorize, and document the most prevalent PCB defect types encountered in electronics repair based on their characteristic visual signatures suitable for automated detection.	Phase 1: Preliminary Study	Conduct comprehensive literature review on PCB defect detection methodologies.	Systematic literature review.	Literature review documentation.
	Phase 2: Knowledge Acquisition	Analyse existing PCB defect datasets.	Dataset analysis (HRIPCB, PKU datasets).	PCB defect taxonomy table.
		Identify common defect types and their visual characteristic.	Defect taxonomy development.	Visual feature characterisation document.
		Consult repair technicians (if feasible) to validate defect relevance.	Expert consultation.	Knowledge gaps identification.

RO2: To design and develop CNN-based detection models using YOLOv12 architecture variants (Nano, Small, Medium), conducting comparative performance evaluation to identify the optimal model for PCB defect detection in repair contexts.	Phase 3: Data Collection & Preparation	Collect and prepare PCB defect datasets.	Data preprocessing & augmentation.	Annotated PCB defect dataset.
	Phase 4: System Design	Design CIRCA system architecture.	Transfer learning.	Trained YOLOv12 models (3 variants).
	Phase 5: System Development	Implement YOLOv12 variants (Nano, Small, Medium).	YOLOv12 implementation (PyTorch/TensorFlow).	Model performance comparison report.
	Phase 6: Testing & Evaluation	Train models on prepared datasets.	Model training & optimisation.	Optimal model selection documentation.
		Evaluate and compare model performance.	Comparative performance analysis.	Detection engine module.
		Identify optimal architecture.	Benchmark metrics (mAP, precision, recall, F1-score, inference speed)	

RO3: To develop the CIRCA (Circuit Inspection and Recognition using Convolutional Architectures) system prototype with a user-friendly interface and evaluate its performance through empirical testing using benchmark metrics (precision, recall, F1-score, and mAP).	Phase 4: System Design	Design system architecture (3-layer: image acquisition, detection engine, UI).	System architecture design.	CIRCA system prototype.
	Phase 5: System Development	Develop user interface for technicians.	User interface development.	User-friendly interface.
	Phase 6: Testing & Evaluation	Integrate all system components.	System integration.	System architecture documentation.
	Phase 7: Documentation	Conduct empirical performance testing.	Empirical testing & validation.	Performance evaluation report (mAP, precision, recall, F1-score).
		Pilot deployment (if feasible).	Usability testing.	User feedback analysis.
		Gather user feedback.	Error analysis.	Implementation challenges documentation.
		Document challenges and limitations.	Technical documentation.	Final thesis report. Code repository

Table 2: Methodology Table

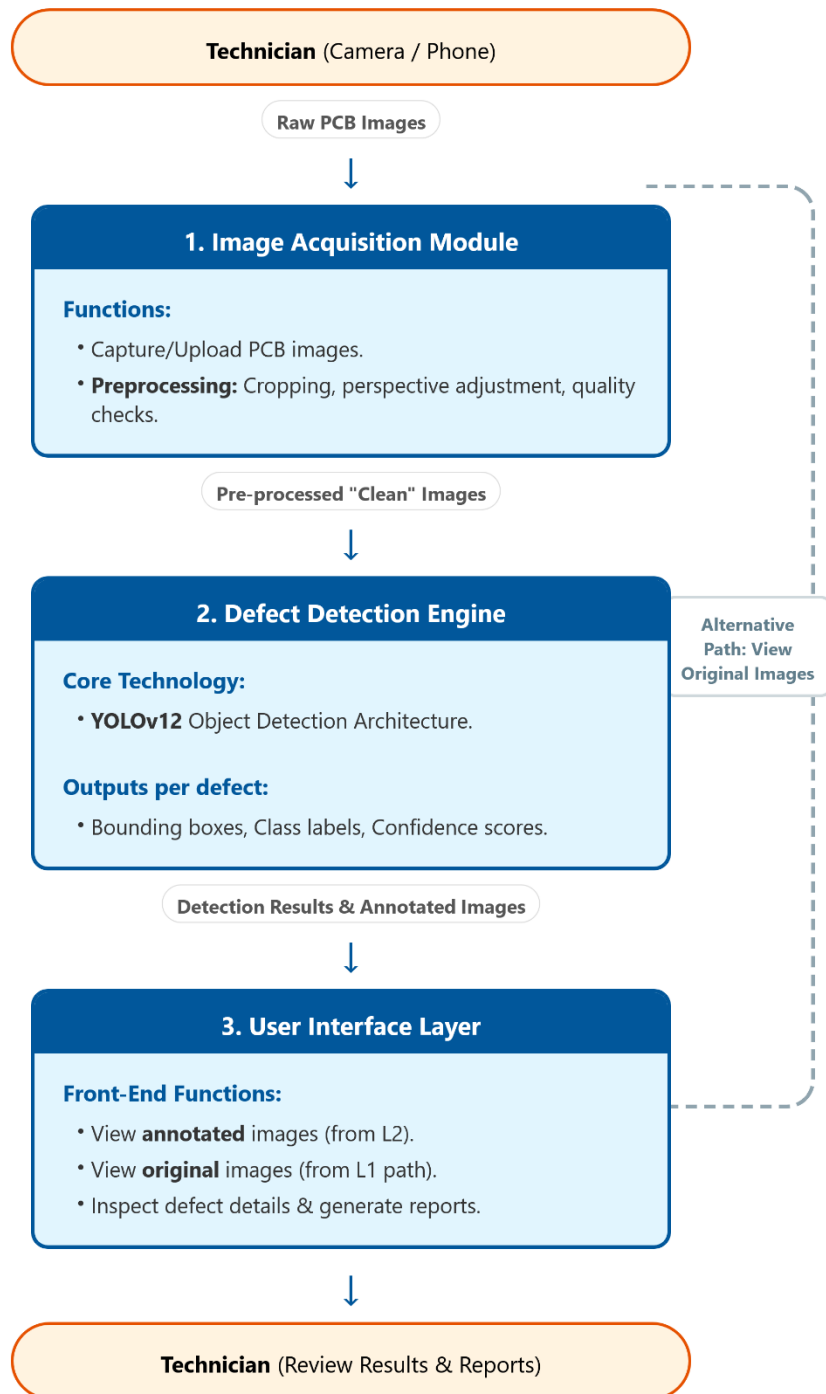
This table serves as the core reference for implementation, ensuring that the preliminary study, data preparation, system design and development, evaluation, and documentation all contribute directly to the defined objectives

3.4 System Architecture

The CIRCA system is designed as a **three-layer architecture** that supports practical deployment in electronics repair environments.

- The **Image Acquisition Module** enables technicians to capture or upload PCB images using standard cameras or smartphones, and performs basic preprocessing such as cropping, perspective adjustment, and quality checking to ensure that images are suitable for defect detection.
- The **Defect Detection Engine** implements the YOLOv12 object detection architecture, which processes the pre-processed images and outputs bounding boxes, defect class labels, and confidence scores for each detected PCB defect.
- The **User Interface Layer** provides an intuitive front-end where technicians can view original and annotated images, inspect defect details, and generate inspection reports without needing to understand the underlying machine learning algorithms.

CIRCA System: Three-Layer Architecture



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